TEXORA · LECTURE SHEET

Statically Determinate Beams

Reactions from equilibrium alone, then the shear-force and bending-moment diagrams that show what the beam feels inside: the standard cases — central and eccentric point loads, uniform load, cantilever — and the rules that let you sketch any diagram before you calculate it.

ME-3210 · SOLID MECHANICS AND MACHINE DESIGN · TOPIC 2 · LEVEL 3 TERM 2

1. Determinate supports and reactions

A beam is **statically determinate** when its reactions follow from the three equilibrium equations alone ($\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M = 0$): the simply supported beam (pin + roller, 3 reactions), the cantilever (fixed end: force, force, moment), and compound beams with internal hinges adding equations. Supports in the mill: a line shaft on two bearing pedestals is a simply supported beam; a bracket or jib is a cantilever; an overhanging shaft end is both ideas at once.

2. Shear force and bending moment — sign discipline

Cut the beam at x and balance the left segment: the internal **shear force** V is the algebraic sum of vertical forces to the left (upward forces positive); the **bending moment** M is the sum of moments to the left about the cut, positive when it sags the beam (compression on top). Three differential relations run the whole subject:

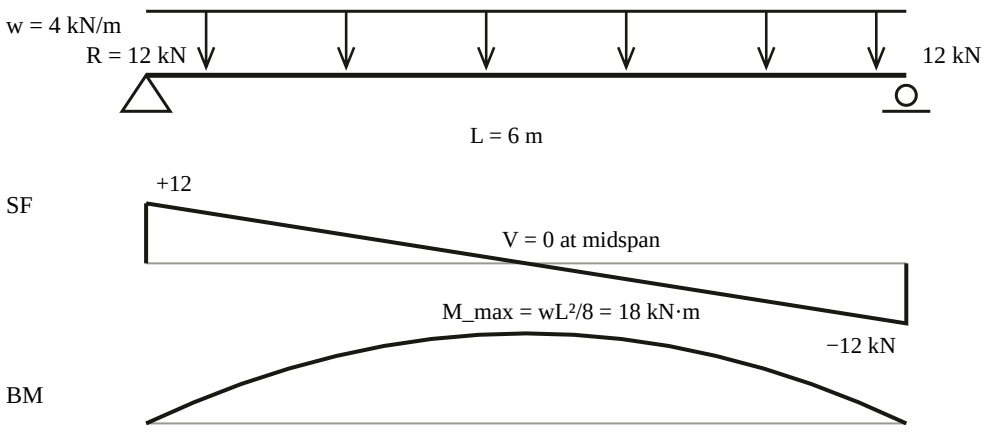
$$dV/dx = -w, \quad dM/dx = V, \quad \text{so M is maximum where V crosses zero}$$

Practical grammar: a point load makes V jump vertically and M change slope; a uniform load makes V slope linearly and M curve as a parabola; no load means V constant and M a straight line; a point moment makes M jump. Learn the grammar and most diagrams sketch themselves.

3. The standard cases

CASE	REACTIONS	MAXIMUM MOMENT
Simply supported, central point load P, span L	P/2 each	PL/4 at centre; SF steps +P/2 → −P/2

Simply supported, point load P at distance a from left (b from right)	R_left = Pb/L, R_right = Pa/L	Pab/L under the load
Simply supported, UDL w over whole span	wL/2 each	wL²/8 at centre; SF zero at centre
Cantilever, end load P, length L	Shear P, moment PL at wall	PL at the fixed end (hogging)
Cantilever, UDL w	Shear wL, moment wL²/2 at wall	wL²/2 at the fixed end



(SF and BM drawn to the same horizontal scale as the beam, on a common vertical alignment)

Figure 1 — Simply supported beam, L = 6 m under UDL w = 4 kN/m: loading, shear-force diagram (linear, +12 → −12 kN, zero at midspan) and bending-moment diagram (parabola, peak 18 kN·m at midspan).

WORKED EXAMPLE — THE SAME NUMBERS, THREE WAYS

UDL case (figured): total load = 4 × 6 = 24 kN, so R = 12 kN each. SF at distance x: V = 12 − 4x, zero at x = 3 m. M there = 12 × 3 − 4 × 3 × 1.5 = 36 − 18 = **18 kN·m** = wL²/8 ✓.

Central point load: 12 kN at midspan of the same beam: R = 6 kN each, M_max = PL/4 = 12 × 6/4 = **18 kN·m** — the same peak from half the total load: concentration is expensive.

Eccentric load: 12 kN at a = 2 m from the left (b = 4 m): R_left = Pb/L = 12 × 4/6 = 8 kN, R_right = 4 kN; M under load = Pab/L = 12 × 2 × 4/6 = **16 kN·m**, and SF jumps from +8 to −4 at the load point.

4. Cantilevers and overhangs

The cantilever is solved from the free end backwards — no reactions need solving at all: shear accumulates the loads, moment accumulates shear × distance, and everything peaks at the wall. A 3 m cantilever with a 6 kN tip load carries constant SF of 6 kN and a wall moment of 6 × 3 = **18 kN·m** (hogging — tension on top, the mirror of sagging). Overhangs put moments at supports: an overhanging load beyond a support creates a negative (hogging) moment there and can lift the far reaction negative — always check for uplift, because a pedestal that expected load may get none, or worse, tension it was never anchored for.

5. Key points

- Equilibrium alone fixes determinate reactions; sections then give V and M anywhere.
- $dV/dx = -w$ and $dM/dx = V$: UDL → linear SF, parabolic BM; M peaks where V crosses zero.
- Standard maxima: $PL/4$ (central load), Pab/L (eccentric), $wL^2/8$ (UDL), PL and $wL^2/2$ (cantilever tip and UDL).
- Point loads step the SF diagram; concentrated moments step the BM diagram.
- Hogging (support/overhang/cantilever moments) puts tension on top — reinforcement and section choice must follow the sign.

6. Practice

Answer in your own words; brief model answers follow.

1. A simply supported beam of 5 m carries a 20 kN load 1 m from the left support. Find the reactions and the moment under the load.
2. A cantilever 2.5 m long carries UDL 3 kN/m. Find the wall shear and moment.
3. A simply supported 8 m beam carries UDL 5 kN/m over its left half only. Find the left reaction. (Hint: treat the load's resultant and its position.)
4. At a section of a beam the SF diagram crosses zero while changing from + to −. What is true of the BM at that section, and why?
5. A 4 m simply supported beam carries a central concentrated moment of 20 kN·m (clockwise on the left segment convention — take moments carefully). Find the reactions.
6. Sketch the SF and BM diagrams for a simply supported beam with equal point loads P at both third points; give M_{max} in terms of P and L.

Answer key

1. $R_{right} = Pa/L$ with a = 1, b = 4: by moments about the left, $R_{right} = 20 \times 1/5 = 4$ kN; $R_{left} = 16$ kN. M under load = $R_{left} \times 1 = 16$ kN·m (check: $Pab/L = 20 \times 1 \times 4/5 = 16$ ✓).

2. Total load = 3 × 2.5 = 7.5 kN = wall shear. Wall moment = $wL^2/2$ = 3 × 6.25/2 = 9.375 kN·m hogging.
3. Resultant = 5 × 4 = 20 kN acting at 2 m from the left support (midpoint of the loaded half). Moments about the right support: R_left × 8 = 20 × 6 → R_left = 15 kN; R_right = 5 kN.
4. The bending moment is a local maximum there (for sagging sign): since dM/dx = V, M is stationary where V = 0, and the + to − crossing of V makes it a peak rather than a trough.
5. Only a couple acts, so vertical reactions form an equal-opposite pair: R × 4 = 20 → R = 5 kN up at one support and 5 kN down at the other; SF is a constant −5 or +5 kN across the span (depending on orientation) and BM varies linearly, jumping by 20 kN·m at the moment's position.
6. Reactions P each. SF: +P from left support to first load, 0 between the loads, −P to the right support. BM rises linearly to P × L/3 at the first load, stays constant at PL/3 between loads (pure bending, V = 0), falls symmetrically. M_max = PL/3.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib